

Financial Modeling and Dealing (FN585)

Basic Probability

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States of the World

Definition

Let S be a set of “states of the world.” Let $p_s > 0$ be the probability of each $s \in S$, so

$$\sum_{s \in S} p_s = 1.$$

The pair (S, p) is called a **probability space**.

Given any subset $E \subset S$, we define

$$p(E) \equiv \sum_{s \in E} p_s.$$

Example

For example, let S be the set of all 36 ordered pairs (i, j) where i and j are integers between 1 and 6. Let $p_s = \frac{1}{36}$ for each $s \in S$. We might think of S as the possible displays from a roll of two dice.

A picture of the dice state space is given below:

	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

Example

Each state s corresponds to an entry in the above matrix. Let E be the states in which the first die shows 3. Then

$$E = \{(i, j) : i = 3\} = \{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6)\}.$$

Thus,

$$p(E) = \frac{6}{36} = \frac{1}{6}.$$

Random Variables on S

- A **random variable** X assigns a number X_s to each state s .
- We often interpret X_s as the **payoff from asset** X in state s .
- Example:
 - X_s could represent the **face of the first die**, so $X(i, j) = i$.
 - Y_s could represent the **face of the second die**, so $Y(i, j) = j$.
- **State s determines both X and Y :**
 - X is constant along each **row**.
 - Y is constant down each **column**.
- Define a third random variable Z :
 - $Z_s = X_s + Y_s$ (total rolled value).
 - Z is constant along **southwest-northwest diagonals**.
- If $u : \mathbb{R} \rightarrow \mathbb{R}$ is any function, then:
 - $u(X)$ is a **random variable**: $u(X_s)$.
- **Key takeaway:** Each state s determines the values of all random variables $X, Y, Z, u(X)$.

Picturing Random Variables

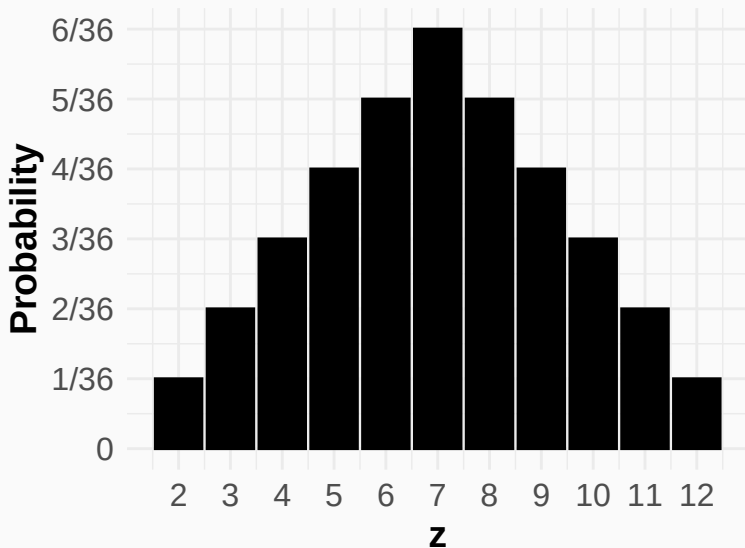
We can explicitly describe how Z depends on the state s by writing the value of Z_s in the cell corresponding to each state s :

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

If the state space is large, the enumeration of all values of Z is hard to see. One simplification is to concentrate on the outcome of Z , $z \in R(Z)$. The *histogram* of Z is the graph associating to each outcome $z \in R(Z)$ its probability $p(\{s \in S : Z_s = z\})$.

Picturing Random Variables

For the sum of two dice, the histogram is:



Picturing Random Variables

- The **highest probability** in the histogram is at $z = 7$.
- Probability **decreases quickly** as z diverges from the center.

Example:

- The **Normal distribution** (bell curve) is defined as:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2}$$

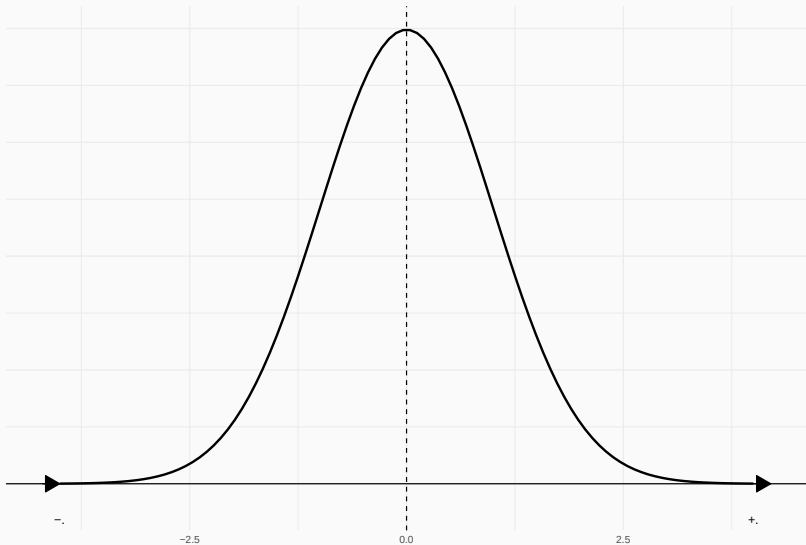
- Falls **doubly exponentially fast** as $|z| \rightarrow \infty$.
 - Known for having a “**thin tail**”.
- By contrast, the **Cauchy distribution**:

$$f(z) = \frac{1}{\pi(1 + z^2)}$$

- Has a “**fat tail**”, meaning probability decays more slowly.

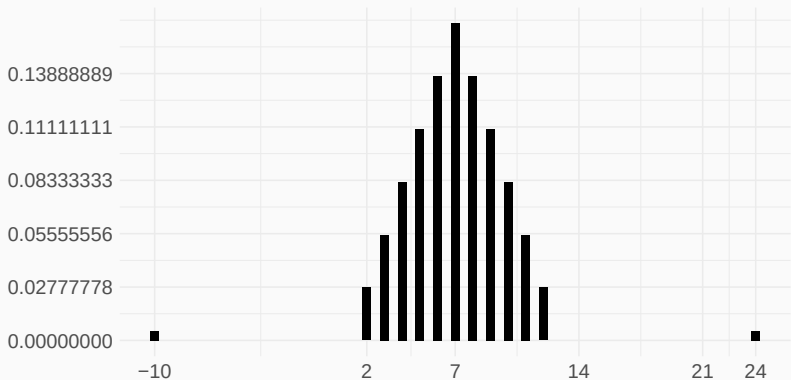
Picturing Random Variables

Standard Normal Distribution



Fat-Tailed Dice Distribution

To give an idea of a fat tail, think of a modified dice in which there were two more outcomes, -10 and 24, each with probability $1/200$. Suppose that the probability of $Z = 2, 3, \dots, 12$ was the old probability multiplied by $99/100$. The histogram then would look roughly like this:



Visualizing a Random Variable as a Vector

- A second way of visualizing a random variable X is as a **vector**.
- By arbitrarily ordering the states from 1 to S , we can represent:

$$X = (X_1, X_2, \dots, X_S)$$

- **Example:** In the dice state space, we can number states from left to right in each row, taking rows from top to bottom, forming a **36-dimensional vector**.

Geometric Properties of X

- Representing X as a vector allows us to use **geometry** to analyze its probabilistic properties.
- Euclidean geometry provides **two key visual insights**:
 - **Vector Lengths**
 - **Angles Between Vectors**
- **Rescaling** the units of X_s makes these geometric ideas applicable to **basic probability concepts**.

Picturing Random Variables

Rescaling in the Geometry of Random Variables

- In the geometry of random variables, we must **account for probabilities** p_s , where $s \in S$.
- The best way to incorporate probabilities is to **rescale the units** so that X_s is replaced by:

$$X_s \sqrt{p_s}$$

- In **Euclidean geometry**, the length of a vector $(X_1, X_2, \dots, X_S)$ is given by:

$$||X|| \equiv \sqrt{\sum_{s=1}^S X_s^2}$$

Graphical Representation of Rescaled Vectors

- After rescaling:

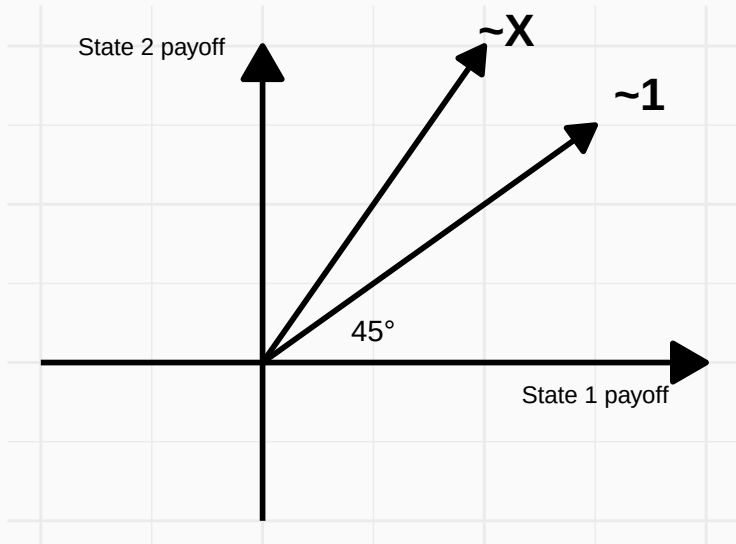
$$\|X\|_p \equiv \sqrt{\sum_{s=1}^S p_s X_s^2}$$

- Consider the **riskless vector** $\tilde{1} \equiv (1, 1, \dots, 1)$.
- The length of the riskless vector:

$$\|\tilde{1}\| = S, \quad \|\tilde{1}\|_p = 1$$

- The following graph shows:
 - **State 1 vs. State 2 Payoff**
 - **Angle 45° between vectors $\tilde{X}$ and $\tilde{1}$**

Graphical Representation of Rescaled Vectors



The Algebra of Random Variables

- **Addition & Multiplication:**

- If X represents the random payoff to one investment and Y represents the payoff to another, then:

$$Z \equiv X + Y, \quad \text{where } Z_s = X_s + Y_s$$

- This represents the **total payoff** if both investments are undertaken.

- **Interaction of Random Variables:**

- If we define multiplication as:

$$Z \equiv XY, \quad \text{where } Z_s = X_s Y_s$$

- This helps us **understand reinforcement**—if both X_s and Y_s are large (either both positive or both negative), Z_s will be large.

The Algebra of Random Variables

- **Scaling by a Constant:**

- If a constant t is applied:

$$Z \equiv tX, \quad \text{where } Z_s = tX_s$$

- This means tX represents the total payoff from **\$t of investments**.

Mean (Expectation)

Definition

The mean or expectation of a random variable X on S is

$$E_p[X] \equiv \sum_{s \in S} p_s X_s.$$

Let $\mathcal{R}(X)$ be the range of the random variable X , that is, $x \in \mathcal{R}(X)$ if and only if $X_s = x$ for some $s \in S$.

Then, we can also write the expectation of X as

$$E_p[X] = \sum_{x \in \mathcal{R}(X)} p\{s : X(s) = x\}x.$$

which can be quickly read off of the histogram.

Expectation Calculation for Dice Rolls

For example, to calculate the expectation of the first die, we should go through every state row by row, writing

$$\begin{aligned} E_p[X] = & \frac{1}{36} \cdot 1 + \frac{1}{36} \cdot 1 + \cdots + \frac{1}{36} \cdot 1 + \frac{1}{36} \cdot 2 + \cdots + \frac{1}{36} \cdot 2 \\ & + \underbrace{\frac{1}{36} \cdot 3 + \cdots + \frac{1}{36} \cdot 3}_{6 \text{ times}} + \underbrace{\frac{1}{36} \cdot 4 + \cdots + \frac{1}{36} \cdot 4}_{6 \text{ times}} \\ & + \underbrace{\frac{1}{36} \cdot 5 + \cdots + \frac{1}{36} \cdot 5}_{6 \text{ times}} + \underbrace{\frac{1}{36} \cdot 6 + \cdots + \frac{1}{36} \cdot 6}_{6 \text{ times}} \end{aligned}$$

Expectation Calculation for Dice Rolls

$$= \frac{6}{36} \cdot 1 + \frac{6}{36} \cdot 2 + \frac{6}{36} \cdot 3 + \frac{6}{36} \cdot 4 + \frac{6}{36} \cdot 5 + \frac{6}{36} \cdot 6$$

$$= \frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 2 + \frac{1}{6} \cdot 3 + \frac{1}{6} \cdot 4 + \frac{1}{6} \cdot 5 + \frac{1}{6} \cdot 6$$

$$= 3.5$$

Expectation Calculation for Dice Rolls

By looking at the histogram of X instead, we could have gone directly to the last two lines in the previous derivation. To calculate the expectation of rolling two dice, we can quickly use the histogram of Z and get

$$\begin{aligned} E_p[Z] &= \frac{1}{36} \cdot 2 + \frac{2}{36} \cdot 3 + \frac{3}{36} \cdot 4 + \frac{4}{36} \cdot 5 + \frac{5}{36} \cdot 6 + \frac{6}{36} \cdot 7 \\ &\quad + \frac{5}{36} \cdot 8 + \frac{4}{36} \cdot 9 + \frac{3}{36} \cdot 10 + \frac{2}{36} \cdot 11 + \frac{1}{36} \cdot 12 \\ &= \frac{252}{36} = 7 \end{aligned}$$

Mean (Expectation)

There is another, geometric way of visualizing the mean of a random variable X . Consider the vector representation of X , and let $\hat{a}$ be the closest point to X on the line $t\mathbf{1}$, in the $\sqrt{p_s}$ -weighted Euclidean metric. Then $E_p(X) = a$, as we prove in the next theorem.

Theorem

Let X be an arbitrary random variable. The solution to the problem

$$\min_a \sum_{s \in S} p_s (X_s - a)^2$$

$$\equiv \min_a \left[p_1(X_1 - a)^2 + p_2(X_2 - a)^2 + \cdots + p_s(X_s - a)^2 \right]$$

is uniquely given by $a = E_p(X)$.

Mean (Expectation)

Proof

If a minimizes the sum, then it must make the derivative zero:

$$\begin{aligned} 0 &= \frac{d}{da} \left(\sum_{s \in S} p_s (X_s - a)^2 \right) \\ &= \sum_{s \in S} 2p_s (X_s - a) \Rightarrow 2 \sum_{s \in S} p_s a = 2 \sum_{s \in S} p_s X_s \end{aligned}$$

Since $\sum_{s \in S} p_s = 1$, we get:

$$a = \sum_{s \in S} p_s a = \sum_{s \in S} p_s X_s \equiv E_p(X)$$

Mean (Expectation)

Theorem Let X , Y and Z be random variables on S with $Z = X + Y$. Then

$$E_p(Z) = E_p(X) + E_p(Y).$$

Proof

$$\sum_{s \in S} p_s(X_s + Y_s) = \sum_{s \in S} p_s X_s + \sum_{s \in S} p_s Y_s. \blacksquare$$

Clearly if $Z = tX$ for some number t , then $E_p(Z) = tE_p(X)$. Thus expectation is a linear operator on random variables.

In our example, let Y be the outcome of the second die $Y(i, j) = j$. Let Z be the sum showing on both dice. Using the last theorem,

$$EZ = 7 = 3.5 + 3.5 = E(X) + E(Y).$$

Note how much simpler this is than calculating the expectation from the histogram...

Averages of Random Variables

The mean is the most important average of a random variable.

But there are others.

Let X be a random variable, and let

$$u : \mathbb{R} \rightarrow \mathbb{R}$$

be an increasing function taking the value of the random variable into another value.

We can always take the expectation of the random variable $u(X)$, but more interesting is:

$$CE_u(X) = u^{-1}(E_p[u(X)])$$

which is often called the **certainty equivalent** of X with respect to u .

Averages of Random Variables

Example: Certainty Equivalent Calculation

If $u(x) = x^2$, and the sample space is $S = \{1, 2\}$ with probabilities $p_1 = p_2 = 1/2$, and $X_s = s$, then:

$$E_p[u(X)] = E_p[X^2] = \frac{1}{2}1^2 + \frac{1}{2}2^2 = 2.5$$

$$u^{-1}(E_p[u(X)]) = CE_u(X) = \sqrt{2.5} = 1.6$$

Notice that the **certainty equivalent is greater than the mean**:

$$E_p[X] = 1.5$$

Certainty Equivalent with Other Transformations

If $u(x) = \sqrt{x}$, then:

$$CE_{\sqrt{x},p}(X) = \sqrt{2.5} = 1.6$$

For logarithmic transformation $u(x) = \log x$:

$$E_p[\log X] = \frac{1}{2} \log 1 + \frac{1}{2} \log 2 = \log \sqrt{1 \times 2} = \log 1.4$$

$$CE_{\log x,p}(X) = \exp(\log \sqrt{1 \times 2}) = 1.4$$

Certainty Equivalent with Other Transformations

For inverse transformation $u(x) = -1/x$:

$$E_p[-X^{-1}] = \frac{1}{2}(1/1) + \frac{1}{2}(1/2) = -0.75$$

$$CE_{-x^{-1},p}(X) = -(1/(-0.75)) = 1.3$$

Averages of Random Variables

Hardy-Littlewood Inequalities

Let X be a random variable, and let $\alpha < \beta$. Then:

$$CE_{x^\alpha, p}(X) < CE_{x^\beta, p}(X)$$

Certainty Equivalents Order

In particular:

$$CE_{-x^{-1}, p}(X) < CE_{\log x, p}(X) < CE_{\sqrt{x}, p}(X) < CE_{x^2, p}(X)$$

The logarithmic certainty equivalent is the **limit** of the certainty equivalent of x^α as $\alpha \rightarrow 0$.

Named Averages & Their Significance

Some of these averages have standard names:

- $CE_{-x^{-1},p}(X)$ is called the **harmonic mean** of X .
- $CE_{\log x,p}(X) = x_{\in S} X_s^{p_s}$ is called the **geometric mean** of X .
- $CE_{x,p}(X) = E_p[X]$ is called the **arithmetic mean** of X .

Thus, the Hardy-Littlewood inequalities include, as a special case, the **famous Greek theorem**:

Theorem of the Means

For any random variable X :

$$\text{Arithmetic Mean} \geq \text{Geometric Mean} \geq \text{Harmonic Mean}$$

unless X is a constant, in which case **all means are identical**.

We now turn to quantifying the “riskiness” of a random variable. This is summarized by how spread out the histogram is. To reduce that to a single number, we define the following:

Definition

The **variance** of a random variable X is defined by:

$$\sigma_X^2 \equiv \text{Var}_p(X) \equiv \sum_{s \in S} p_s (X_s - E_p(X))^2.$$

Variance

Example

The variance of the die is:

$$\begin{aligned} & \frac{6}{36}(1 - 3.5)^2 + \frac{6}{36}(2 - 3.5)^2 + \frac{6}{36}(3 - 3.5)^2 \\ & + \frac{6}{36}(4 - 3.5)^2 + \frac{6}{36}(5 - 3.5)^2 + \frac{6}{36}(6 - 3.5)^2 \\ & = \frac{6}{36}(6.25 + 2.25 + .25 + .25 + 2.25 + 6.25) \\ & = \frac{6}{36}(17.50) = 2.916. \end{aligned}$$

Definition

The **standard deviation** of a random variable X is defined by:

$$\sigma_X \equiv \sqrt{\text{Var}_p(X)} = \sqrt{\sum_{s \in S} p_s (X_s - E_p(X))^2}.$$

The standard deviation of the die is thus $\sigma_X = \sqrt{2.916} \approx 1.7$.

Variance

A random variable with 0 variance has 0 standard deviation and always turns up equal to its expectation. The variance is a measure of the average squared deviation from the mean. From the Hardy-Littlewood inequalities, we know that the standard deviation is always at least as big as the average absolute deviation from the mean:

$$\sqrt{\sum_{s \in S} p_s (X_s - E_p(X))^2} \geq \sum_{s \in S} p_s |X_s - E_p(X)|.$$

example

For example, with the die:

$$\sigma_X = \sqrt{2.916} \approx 1.7.$$

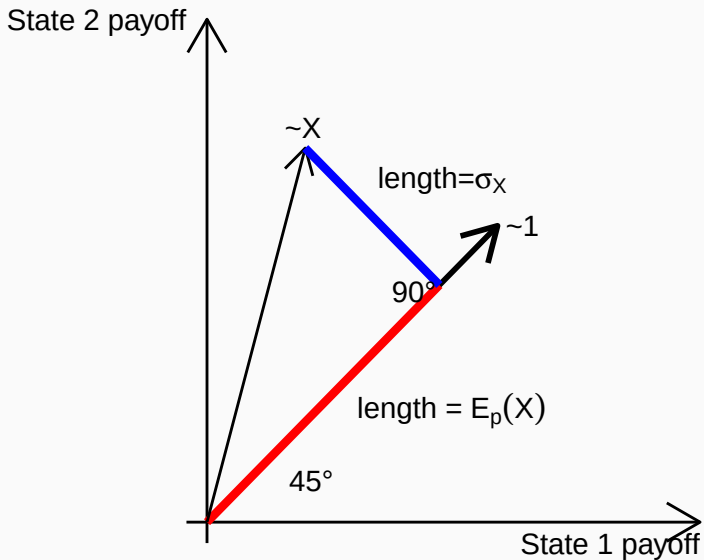
The average absolute deviation is:

The Fame of Standard Deviation

The reason for the fame of the standard deviation (as opposed to expected absolute deviation) has to do with the computational and geometric simplicity of variance, and the fact that normally distributed random variables are defined by their mean and standard deviation.

We can visualize the standard deviation of a random variable X by noting that it is the (p -Euclidean) distance from X to $[E_p(X)]\tilde{1}$.

Variance



Variance

This geometric decomposition of any vector X into its mean and standard deviation is another reason that these two concepts have been singled out as special in probability theory. We write this formally as:

Theorem

For any random variable X ,

$$\begin{aligned} E_p(X^2) &\equiv \sum_{s \in S} p_s X_s^2 = \sum_{s \in S} p_s E_p(X)^2 + \sum_{s \in S} p_s (X_s - E_p(X))^2 \\ &\equiv [E_p(X)]^2 + \text{Var}_p(X). \end{aligned}$$

Proof

$$\begin{aligned}& \sum_{s \in S} p_s (X_s - E_p(X))^2 \\&= \sum_{s \in S} p_s X_s^2 - 2 \sum_{s \in S} p_s X_s E_p(X) + \sum_{s \in S} p_s E_p(X)^2 \\&= \sum_{s \in S} p_s X_s^2 - 2E_p(X)^2 + E_p(X)^2 \\&= \sum_{s \in S} p_s X_s^2 - E_p(X)^2.\end{aligned}$$

Variance

The variance does quantify how risky a random variable is in the following sense. People have defined the “value at risk” as the maximum amount by which X could fall short of its expectation $E_p(X)$ with probability at least 5%. In other words, if $p(\{s \in S : X \leq E(X) - L\}) \geq 0.05$, then the value at risk is at least L . The variance puts a bound on the value at risk.

Theorem

$$L \leq \frac{\sigma_X}{\sqrt{p(\{s \in S : X \leq E(X) - L\})}}$$

In particular,

$$\text{Value at risk} \leq \sigma_X \sqrt{20}.$$

Proof

$$\begin{aligned}\sigma_X^2 &\geq L^2 p(\{s \in S : (X - E(X))^2 \geq L^2\}) \\ &\geq L^2 p(\{s \in S : X - E(X) \leq -L\}). \blacksquare\end{aligned}$$

This bound can be made even tighter for normally distributed random variables. For normally distributed random variables, a loss of $2\sigma_X$ (a “two standard deviation loss”) has probability less than 2.5%.

Theorem For any random variable X , and constants t and a ,
 $Var(tX + a) = t^2 Var(X)$.

Proof

$$\begin{aligned} & \sum_{s \in S} p_s (tX_s + a - [tE_p(X) + a])^2 \\ &= \sum_{s \in S} p_s (tX_s - tE_p(X))^2 \\ &= t^2 \sum_{s \in S} p_s (X_s - E_p(X))^2. \blacksquare \end{aligned}$$

We now consider two or more random variables defined on the same state space S .

Covariance

Definition The covariance of two random variables X and Y is:

$$\sigma_{XY} = \text{Cov}_p(X, Y) \equiv \sum_{s \in S} p_s(X_s - E_p(X))(Y_s - E_p(Y)).$$

Clearly $\text{Var}(X) = \text{Cov}_p(X, X)$ for any random variable X . If $\text{Cov}_p(X, Y) > 0$, the two variables tend to “move together”: when X_s is above its mean, Y_s is usually above its mean. Clearly $\sigma_{XY} = \sigma_{YX}$.

Lemma If $Y \equiv a$ is a constant, then $\text{Cov}_p(X, Y) = 0$, for all random variables X .

Proof

$$\text{Cov}_p(X, Y) = \sum_{s \in S} p_s(X_s - E_p(X))(a - a) = 0. \blacksquare$$

Lemma Let X, Y, Z be random variables. Then

$$\text{Cov}_p(X, Y + Z) = \text{Cov}_p(X, Y) + \text{Cov}_p(X, Z).$$

Proof

$$\begin{aligned} & \sum_{s \in S} p_s (X_s - E_p(X))(Y_s + Z_s - (E_p(Y) + E_p(Z))) \\ &= \sum_{s \in S} p_s (X_s - E_p(X))(Y_s - E_p(Y) + Z_s - E_p(Z)) \\ &= \sum_{s \in S} p_s (X_s - E_p(X))(Y_s - E_p(Y)) + \sum_{s \in S} p_s (X_s - E_p(X))(Z_s - E_p(Z)). \blacksquare \end{aligned}$$

Clearly $\text{Cov}_p(X, tY) = t\text{Cov}_p(X, Y)$ for any number t . Hence, fixing X , $\text{Cov}_p(X, Y)$ is a linear operator on random variables Y .

Covariance

Definition If $\sigma_X > 0$ and $\sigma_Y > 0$ the correlation of X and Y is defined as:

$$\rho_{XY} \equiv \frac{\sigma_{XY}}{\sigma_X \sigma_Y}.$$

Theorem Let X and Y be arbitrary random variables with $\sigma_X > 0$ and $\sigma_Y > 0$. Let x and y be their standard versions. Then $|\rho_{XY}| = |\sigma_{XY}/\sigma_X \sigma_Y| = |\sigma_{xy}| \leq 1$, with equality only if X and Y are perfectly correlated or perfectly negatively correlated.

We can also give an interpretation of the correlation $\rho_{XY} = \sigma_{xy}$. Suppose we must find the line $y = ax$ that best predicts y , given any value of x . What should a be? The answer depends on the punishment criterion for guessing wrong. But if the punishment is the square of the error, then minimizing the expected punishment requires putting $a = \sigma_{xy}$.

Geometric Interpretation of Correlation

We can give two geometric interpretations of the correlation of random variables X and Y .

First, let us put them in their so-called “*standard form*.”

Define the random variables x and y by

$$x_s \equiv \frac{X_s - \bar{X}}{\sigma_X}, \quad y_s \equiv \frac{Y_s - \bar{Y}}{\sigma_Y}.$$

This simply amounts to changing the scale in which the variables are measured.

Note that $E_p(x) = E_p(y) = 0$ and $\sigma_x = \sigma_y = 1$.

Geometric Interpretation of Correlation

Also, note that

$$\rho_{XY} = \sigma_{xy}.$$

The correlation of the random variables X and Y is identical to the covariance between the standardized random variables x and y .

To understand when X and Y are **positively correlated**, **uncorrelated**, or **negatively correlated**, it suffices to understand the correlation of x and y .

Correlation and Quadrants

Since $\sigma_{xy} = \sum_{s \in S} p_s x_s y_s$,

- If the points (x_s, y_s) are **simultaneously positive** (first quadrant) or **simultaneously negative** (third quadrant), the correlation is **positive**.
- If the points lie on the **45° line** $y = x$, then x and y (and therefore X and Y) are **perfectly correlated**.

$$\sigma_{xy} = \sum_{s \in S} p_s x_s y_s = \sum_{s \in S} p_s x_s^2 = \sigma_x^2 = 1.$$

Negative Correlation

- If most points (x_s, y_s) are in the **2nd and 4th quadrants**, the correlation is **negative** (when one goes up, the other tends to go down).
- If points lie on the **line** $y = -x$, then x and y (and therefore X and Y) are **perfectly negatively correlated**.

$$\sigma_{xy} = \sum_{s \in S} p_s x_s y_s = \sum_{s \in S} -p_s x_s^2 = -\sigma_x^2 = -1.$$

Interpreting Correlation for In-Between Cases

- When the points (x_s, y_s) are **not on either diagonal**, we analyze their closest projection onto the line $y = x$.
- The closest point to (x_s, y_s) on $y = x$ is:

$$\left(\frac{x_s + y_s}{2}, \frac{x_s + y_s}{2} \right).$$

- The difference:

$$(x_s, y_s) - \left(\frac{x_s + y_s}{2}, \frac{x_s + y_s}{2} \right) = \left(\frac{x_s - y_s}{2}, \frac{y_s - x_s}{2} \right)$$

is **orthogonal** to the line $y = x$, confirming its role as the closest point.

Squaring the Distance and Average Squared Difference

- Squaring the distance:

$$(x_s - y_s)^2/4 + (y_s - x_s)^2/4 = \frac{2x_s^2 + 2y_s^2 - 4x_sy_s}{4}.$$

- Averaging over all $s \in S$, weighted by probabilities p_s :

$$d^2 \equiv \sum_{s \in S} p_s \left(\frac{2x_s^2 + 2y_s^2 - 4x_sy_s}{4} \right) = \frac{1}{2} + \frac{1}{2} - \sum_{s \in S} p_s x_s y_s = 1 - \sigma_{xy}.$$

Interpreting the Result

- Since d^2 is always **nonnegative**, we conclude:

$$d^2 \geq 0 \quad \text{and} \quad \sigma_{xy} \leq 1.$$

- Replacing x with $-x$, we obtain:

$$-1 \leq \sigma_{xy} \leq 1.$$

- Since $d^2 > 0$ unless $x_s = y_s$ for all s , we interpret:

$$\sigma_{xy} = 1 - d^2.$$

- This confirms that σ_{xy} is **bounded between -1 and 1**, attaining the extreme values only when x and y are **perfectly correlated or anti-correlated**.

Theorem Let x and y be arbitrary random variables with $\sigma_x^2 = 1$.
Then the solution to

$$\min_a \sum_{s \in S} p_s (y_s - ax_s)^2$$

occurs at $a = \sum_{s \in S} p_s x_s y_s$.

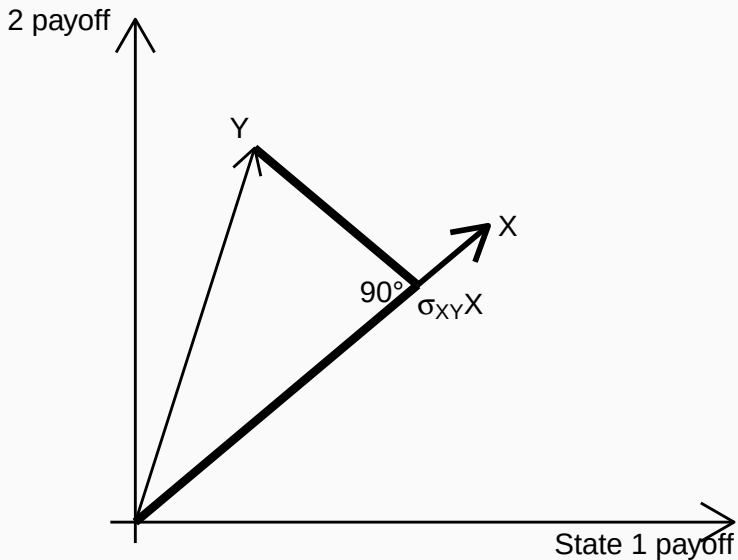
Proof

$$\begin{aligned} 0 &= \frac{d}{da} \sum_{s \in S} p_s (y_s - ax_s)^2 \\ &= 2 \sum_{s \in S} p_s (y_s - ax_s)(-x_s) \end{aligned}$$

So

$$\sum_{s \in S} p_s x_s y_s = a \sum_{s \in S} p_s x_s^2 = a. \blacksquare$$

Covariance



Independence

Definition The random variables X and Y are called uncorrelated if $\text{Cov}_p(X, Y) = 0$.

Definition The random variables X and Y are independent if for all numbers a and b ,

$$\sum_{s \in X^{-1}(a) \cap Y^{-1}(b)} p_s = \text{Prob}(X = a \& Y = b) = [\text{Prob}(X = a)][\text{Prob}(Y = b)]. \blacksquare$$

The random variables are independent if and only if knowing which value Y takes on provides no information about which value X takes on and vice versa. The best guess of Y_s given X_s is $E_p Y$. Hence from the second interpretation of correlation we get:

Theorem If X and Y are independent, then $\text{Cov}_p(X, Y) = 0$.

Theorem If X and Y are independent, then
 $Var(X + Y) = Var(X) + Var(Y)$.

Proof

$$\begin{aligned} Var(X + Y) &= Cov_p(X + Y, X + Y) \\ &= Cov_p(X + Y, X) + Cov_p(X + Y, Y) \\ &= Cov_p(X, X) + Cov_p(Y, X) + Cov_p(X, Y) + Cov_p(Y, Y) \\ &= Cov_p(X, X) + 0 + 0 + Cov_p(Y, Y). \blacksquare \end{aligned}$$

Computation of $\text{Var}_p(Z)$

In the dice example, X and Y are independent. The variance of each die was calculated as **2.916**. Hence, the variance of their sum Z is:

$$\text{Var}_p(Z) = \text{Cov}_p(X + Y, X + Y) = \text{Var}_p(X) + \text{Var}_p(Y) = 5.833.$$

Independence

Computation of $\text{Var}_p(Z)$

$$\text{Var}_p(Z) = \frac{1}{36}(2-7)^2 + \frac{2}{36}(3-7)^2 + \frac{3}{36}(4-7)^2 + \frac{4}{36}(5-7)^2 + \frac{5}{36}(6-7)^2 +$$

$$\frac{6}{36}(7-7)^2 + \frac{5}{36}(8-7)^2 + \frac{4}{36}(9-7)^2 + \frac{3}{36}(10-7)^2 + \frac{2}{36}(11-7)^2 + \frac{1}{36}(12-7)^2$$

$$= \frac{25 + 32 + 27 + 16 + 5 + 0 + 5 + 16 + 27 + 32 + 25}{36} = \frac{210}{36}.$$

Stock Price Example

If X represents stock price changes on day 1 and Y on day 2, then $Z = X + Y$ represents the total two-day change.

- Expected two-day change $E_p(Z) = E_p(X) + E_p(Y)$.
- If changes are **independent**, then:

$$\text{Var}_p(Z) = \text{Var}_p(X) + \text{Var}_p(Y).$$

- More generally, annual expected price changes are the sum of daily expected changes, and **annual variance** is the sum of **daily variances**.

Covariance Computation

Since X and Y are independent:

$$\text{Cov}_p(X, Z) = \text{Cov}_p(X, X + Y) = \text{Cov}_p(X, X) + \text{Cov}_p(X, Y).$$

$$= \text{Var}_p(X) + 0 = 2.916.$$

Sampling from a State Space

- We have a **random variable** X , but we do not know its exact values X_s .
- However, we can **sample from** S .
- Let $\bar{S}$ be a **sampled state space**, generated by **repeatedly** drawing $s \in S$ according to probability p_s .
- For each s drawn, we **observe the values** of all random variables at s .
- The **probability on** $\bar{S}$ is:

$$p_{\bar{s}} = \frac{1}{\#\bar{S}}, \quad \forall \bar{s} \in \bar{S}.$$

Faithfulness of Sampled Space

- Is $\bar{S}$ a **faithful representation** of S ?
 - If $\#\bar{S} < \#S$, important possibilities might be **missing**.
 - If $\#\bar{S} \gg \#S$, the frequency of a state being chosen **approaches** p_s .
- But **convergence is slow!**

Example:

- Suppose S consists of two points: s_1 and s_2 .
- Let $p_1 = 999/1000$ and $p_2 = 1/1000$.
- If **1000 points** are drawn:
 - The probability that s_2 **never appears** is:

$$\left(1 - \frac{1}{1000}\right)^{1000} = \frac{1}{e} \approx 36\%.$$

Correlation Calculation

- Even with **2000 samples**, the probability that s_2 is never chosen is:

$$\frac{1}{e^2} \approx 13\%.$$

- **Implication:** Rare but important events might be **missed in a sample space**.

$$\rho_{XZ} = \frac{\sigma_X \sigma_Z}{\sigma_X \sigma_Z} = \frac{2.916}{\sqrt{2.916} \times \sqrt{2.916}} = \frac{1}{\sqrt{2}} \approx 0.7.$$